AI Ecosystem and Discovery in AI Network

Separation of General Discovery and Agent Protocols in AI Ecosystems

draft-akhavain-moussa-ai-network
draft-am-layered-ai-discovery-architecture

Arashmid Akhavain arashmid.akhavain@huawei.com
Hesham Moussa hesham.moussa@huawei.com

Huawei Technologies Canada

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Background: AI Network Draft

Proposes a consumer-provider overlay network architecture supporting:

- Agentic interactions
- AI training workflows
- AI inference workflows
- Discovery of different entities within AI ecosystem
- Cross-domain interoperability

Covers:

- AI network architecture for training, inference, and AI workflows
- Discovery of datasets, models, tasks, resources, and agents
- Cross-domain interoperability and agentic interactions

The proposed architecture serves to highlight areas that could benefit from standardised protocols and mechanisms, rather than being the focus of standardisation itself.

AI Network architecture, building blocks, and protocols

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AI service provider community

- Data providers
- Model providers
- Agents providers
- Resource providers
- Tools ???

AI client community

- Data clients
- Model clients
- Agent clients
- Resource clients
- Tools ???

AI Network Protocols CP, DP, OAM

AI Network Core

Others???

- X-RCE
- X-RRT
- S-FAM
- MDO
- TAG

AI Network Core Protocols CP, DP, OAM

- Tools Mgmt. ???
- Registration & Authentication
- Discovery & advertisement
- Billing/Accounting
- Trust & Reputation Mgmt.
- Upgrade Mgmt.
- Validation

AI Network Enablers

Communications/Networking

- Mobile network
- Internet
- NTN
- WiFi

Location

- GPS
- Mobile network
- Sensors

Others ???

Observations

Common work items in AI ecosystem such as “discovery” can span multiple object categories and types:

- Agents
- Tasks
- Datasets
- Models
- Resources

Questions:

How to structure standardisation work around these work items?

How should IETF standardisation divide responsibilities?

Agent ∈ AI Ecosystem

Agent interactions	=	agent discovery	+	agent communications.
Training interactions	=	data discovery	+	training process.
Resource interactions	=	resource discovery	+	resource employment.
Task interactions	=	task discovery	+	task assignment.

…

Taking agents as an example, two separable concerns appear:

1. Discovery substrate: Mechanisms for advertising, indexing, and querying discoverable objects.

2. Agent interaction: Protocols governing agent communications and session management **after discovery**

The proposal is to separate these concerns into distinct WGs.

Discovery as a General Substrate

General discovery addresses:

- Object announcement and advertisement
- Metadata schema and header definition framework
- Query and filtering semantics
- Extensible object typing
- Subscription to discovery service

It answers:

What objects exist, and how they are discovered?

Agent cards are one object type within this model.

Agent Protocol Work

Agent-related work includes:

- Capability expression semantics
- Invocation and interaction protocols
- Negotiation workflows
- Multi-agent coordination
- Communication protocols including multi modality
- Session management

It answers:

How agents engage with other agents and the rest of the AI ecosystems (e.g. tools) after being discovered.

Layered Model Proposal

Layer 1 — General AI Discovery WG

- Substrate for discovering all object types
- Metadata schema and headers, query, advertisement, and type extensibility
- Includes agent discovery, excludes agent behavioural semantics

Layer 2 — Agent Protocol WG

- Protocols for interactions between discovered agents and tools
- Communications, session management, multi modality, invocation, negotiation, coordination, etc.

```
+-----------------------------+
| Agent Protocols WG          |
|-----------------------------| Other WGs
| Capability semantic         | with
| Invocation protocol         | discoverable
| Negotiation & coordination  | objects
| etc.                        | . . .
+-----------------------------------------+
|         General AI Discovery WG         |
|-----------------------------------------|
| Object registration                     |
| Metadata schema framework               |
| Query and filtering                     |
| Object transport and delivery           |
+-----------------------------------------+
```

Discoverable Objects

Discovery with extensible types:

- Uniform query and advertisement model
- Typed metadata per object category

```
+-------------------------+
|  Discoverable Objects   |
+-------------------------+
   /     |        |      \
  /      |        |       \
[Task] [Agent] [Model] [Dataset]
```

No separate registry for agents

Boundary Clarification

Layer	In Scope	Non-Goals
Agent WG	• Agent capability description semantics • Agent invocation protocol(s) • Negotiation workflows • Multi-agent coordination • Communication protocols, session management, multi-modality	• Designing general discovery registries, indexing, etc. • Defining cross-type search infrastructure • Standardising dataset/model discovery mechanisms • Standardising general metadata query frameworks • Trust, registration, authorisation, privacy, security, HIL????
Discovery WG	• General discovery vehicle • Generic object model for AI ecosystem discovery • Header definition framework, metadata schema (extensible typed header) • Query language or API specification • Scalability	• Defining agent invocation semantics • Defining negotiation protocols • Defining task allocation algorithms • Defining agent behavioural requirements • Registration, authorisation, privacy, security

WG responsibilities

Agent WG defines

- How agents interwork, collaborate, and communicate
- How interactions occur after discovery

Discovery WG defines

- How objects are described
- How objects are located

This maintains separation between discovery infrastructure and agent runtime protocols

Rationale for Separation

1. Maintaining distinct abstraction layers. Aligns well with IETF layering principles
2. Retaining focus on mission specific protocols and mechanisms
3. Reduced charter ambiguity and scope creep avoidance
4. Promoting parallel progress across working groups
5. Maximising reusability of discovery infrastructure for different object types

Principles:

- The split follows common layering principles used in IETF protocol and top down design.

- Each WG has well-defined responsibilities; one consumes outputs from the other.

Discovery in AI Network

- Agents can be treated as a typed object within a general discovery substrate.
- Discovery and interaction are distinct architectural layers.

```
AI Ecosystem Objects
+--------------------------------+
| Tasks | Models | Data | Agents |
+--------------------------------+
            |
            v
+----------------------------+
| General Discovery WG       |
| Metadata, Typing, Query    |
+----------------------------+
            |
            v
+----------------------------+
| Agent Protocol WG          |
| Invocation, Negotiation    |
| Coordination, Trust        |
+----------------------------+
```

Employing common discovery vehicle to locate discoverable entities

Final words

Objectives: Separate the common from the specific

- Common cross-entity requirements → Address them in broad, ecosystem-wide venues to enable interoperability and consistency.
- Entity-specific work items → Address them in specialised venues with sharper technical focus.

Advantages of the proposed work division:

- Standardisation can proceed in parallel with clearly defined scopes.
- Explicit non-goals helps reducing overlap and ambiguity.
  - General discovery WG handles all discoverable objects, including agents.
  - Agent protocol WG handles behavioural semantics after discovery.
- Separation clarifies scope, prevents overlapping goals, and allows modular standardisation.
- Layering aligns with IETF best practices for protocol development.

Thank You.

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Potential concerns and responses

Potential concerns and responses:

Agents are just objects – why separate WGs?

- Conceptually, yes: agents are objects.
- Practically, agents are active participants with runtime behaviour, negotiation, multi modal interactions.
- Discovery substrate handles object advertisement, query, resolution, and service subscription.
- Agent specific WG handles protocol semantics for communications, session management, and agent card definition.

Won’t this create dependency issues?

- Dependency is unidirectional: Agent protocols consume discovery outputs.
- Discovery layer is independent, stable, and reusable across object types.
- No need for the discovery WG to standardise agent behaviour.
- Allows parallel progress without blocking.

Potential concerns and responses:

Couldn’t a single WG handle both discovery and agent protocols?

- Combining both risks:
  - Overly broad charter
  - Slow consensus on multiple problem spaces
  - Scope creep

- Separate WGs provide:
  - Clear deliverables
  - Modular standards
  - Independent evolution of discovery and agent protocol layers

Does this fragment the ecosystem?

- No. Discovery is unified; agents are treated as one object type.
- Agent protocol WG builds on the same substrate, no duplicate registries.
- Enables cross-object searches while handling agent interactions in a separate protocol layer.